

David Manuel

Staff Full Stack Engineer

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PROFESSIONAL SUMMARY

Staff level Full Stack Software Engineer with over 10 years of experience designing and architecting high-performance enterprise platforms, agentic AI systems, and dynamic canvas-based interfaces. Expert in React, TypeScript, micro-frontends, real-time architectures (WebSockets, SSE), dynamic state management, and full-stack API orchestration (BFF pattern, REST, GraphQL, Node.js). Proven track record of owning frontend direction, driving micro-frontend strategies across distributed engineering teams, and optimizing UI performance for live workflow engines. Seeking to lead frontend and UI architecture as Staff Full Stack Engineer at Wand Synthesis AI Inc.

CORE COMPETENCIES

Frontend Architecture & Engineering: React, TypeScript, Next.js, Micro-Frontends, JavaScript (ES6+), HTML5, CSS3, Tailwind CSS, Material UI, UI/UX Systems, SSR, Performance Optimization

Canvas & Interactive UI: Dynamic Canvas/Node-Based UI, Real-Time State Management (Zustand, Redux), Dynamic Forms & Workflow Builders, WebSockets, Server-Sent Events (SSE)

Backend & API Orchestration: Node.js, Express.js, NestJS, Backend-For-Frontend (BFF), RESTful APIs, GraphQL, gRPC, WebSockets, Python (FastAPI, Flask), .NET Core, Java Spring Boot

Data, Cloud & DevOps: PostgreSQL, MongoDB, Redis, Elasticsearch, Pinecone, AWS (EKS, ECS, Lambda), Docker, Kubernetes, CI/CD (GitHub Actions, Azure DevOps), System Architecture

PROFESSIONAL EXPERIENCE

Staff Full Stack Engineer

Dec 2022 – Present | Rochester, NY

Netsmartz

- Architect and own the end-to-end frontend architecture for an Enterprise AI Copilot Studio using React, TypeScript, and Micro-Frontends, enabling cross-functional engineering teams to integrate UI modules without breaking state or UI patterns.
- Designed and built real-time, canvas-based dynamic workflow visualizers with live UI status updates streaming over WebSockets and Server-Sent Events (SSE), reflecting real-time agent execution states.
- Architected a robust Backend-For-Frontend (BFF) layer in Node.js / Express, unifying REST and GraphQL services to orchestrate complex AI/LLM multi-agent workflows (OpenAI, Claude, LangChain, RAG) directly into the UI state layer.
- Reduced median API and frontend rendering latency from ~850ms to 380ms (55% performance boost) by implementing smart Redis caching, client-side virtualized rendering, and optimized async UI updates.
- Spearheaded dynamic state management using Zustand and Redux, engineering responsive, node-based interactive components that handle complex, live-updating data flows under heavy load.
- Established company-wide frontend standards, automated CI/CD deployment pipelines, code review norms, and reusable component libraries to uphold code quality across multi-team deliverables.

Senior Full Stack Software Engineer

Feb 2020 – Nov 2022 | Chicago, IL

Tempus AI

- Engineered complex, data-heavy dashboard interfaces and workflow builders using React, TypeScript, and REST/GraphQL APIs, enabling clinicians to configure automated analytical pipelines.
- Built real-time UI notifications and data streams using WebSockets, allowing users to monitor live data processing workflows across distributed cloud microservices.
- Reduced p95 frontend rendering and backend API latency from 620ms to 470ms (24% improvement) through efficient database indexing, React rendering optimizations, and async task execution.
- Implemented strict enterprise authentication and access controls (OAuth2, JWT, RBAC, field-level encryption, audit logging) meeting HIPAA and strict security standards.
- Containerized and deployed frontend applications and BFF services across AWS and Google Cloud using Docker, Kubernetes, and CI/CD pipelines.

Software Engineer

Dec 2015 – Jan 2020

Talmate

- Developed interactive full-stack management dashboards using React, JavaScript, Node.js, and Java Spring Boot for enterprise workflow automation.
- Reduced average workflow execution and UI update times from 12 minutes to under 6 minutes (50% performance improvement) via optimized query tuning, Redis caching, and reactive state updates.
- Automated CI/CD integration and deployment workflows using Docker, Kubernetes, GitHub Actions, and Azure DevOps.
- Partnered closely with product management and UX design to transform complex requirements into highly usable enterprise UI components.

EDUCATION

Bachelor's Degree in Computer Science

Aug 2011 – May 2015 | Houston, TX

University of Houston